HANDWRITTEN NOTES

COMPLEX VARIABLES, PROBABILITY AND STATISTICS

Subject Code : 23A54403

UNIT - 1

Complex Variable - Differentiation

Topics covered in this unit :

- Introduction to functions of a complex variable
- Concept of limit and continuity
- Differentiation, Cauchy-Riemann equations
- Analytic functions, harmonic functions
- Finding harmonic conjugate
- Construction of analytic function: Milne-Thomson method

(Key terms are written in red , headings in purple/green, formulas on yellow.)

1. Complex Numbers - Quick Recap

$$z = x + iy \;,\; i^2 = -1 \;,\; x = Re(z) \;,\; y = Im(z)$$

Item	Formula
Conjugate	$\bar{z} = x - iy$
Modulus	$\lvert z\rvert = r = \sqrt{x^2 + y^2}$
Argument	$\theta = \arg z = \tan^{-1}(y/x)$ (principal value $-\pi < \theta \le \pi$)
Polar form	$z = r(\cos\theta + i\sin\theta) = r e^{i\theta}$
Product / quotient	$z_1 z_2 = r_1 r_2 e^{i(\theta_1+\theta_2)}$; $z_1/z_2 = (r_1/r_2) e^{i(\theta_1-\theta_2)}$
De Moivre	$(\cos t + i\sin t)^n = \cos nt + i\sin nt$
Euler	$e^{it} = \cos t + i\sin t$; $\cos t = (e^{it}+e^{-it})/2$; $\sin t = (e^{it}-e^{-it})/2i$
Triangle inequality	$\lvert z_1 + z_2\rvert \le \lvert z_1\rvert + \lvert z_2\rvert$

Useful points

- $z\,\bar{z} = \lvert z\rvert^2$; $1/z = \bar{z} / \lvert z\rvert^2$
- Argand plane: x-axis = real axis, y-axis = imaginary axis
- Neighbourhood of z_0: $\lvert z - z_0\rvert < \delta$ (open disc); region = open connected set
- $\lvert z - a\rvert = r$ is a circle ; $\lvert z - a\rvert < r$ is its interior

2. Functions of a Complex Variable

2.1 Definition

If to each complex number $z = x + iy$ of a set S there corresponds one (or more) complex number $w = u + iv$, then w is a function of z : $w = f(z) = u(x, y) + i\,v(x, y)$, where u and v are real functions of x and y.

- Single valued: one w for each z (e.g. z^2, e^z). Multi-valued: more than

one w (e.g. z^(1/2), log z).

Examples

f(z)	u(x,y)	v(x,y)
z^2	x^2 - y^2	2xy
z^3	x^3 - 3xy^2	3x^2 y - y^3
e^z	e^x cos y	e^x sin y
sin z	sin x cosh y	cos x sinh y
cos z	cos x cosh y	- sin x sinh y
1/z	x/(x^2+y^2)	-y/(x^2+y^2)
z-bar	x	-y
|z|^2	x^2 + y^2	0

2.2 Elementary functions

Function	Definition / property
Exponential	e^z = e^x (cos y + i sin y); periodic with period 2 pi i; e^z != 0
Trigonometric	sin z = (e^(iz) - e^(-iz))/2i ; cos z = (e^(iz) + e^(-iz))/2
Hyperbolic	sinh z = (e^z - e^-z)/2 ; cosh z = (e^z + e^-z)/2 ; sin(iz) = i sinh z
Logarithm	log z = log r + i (theta + 2 n pi); principal value Log z = log r + i theta
Power	z^a = e^(a log z)

Remember

- sin^2 z + cos^2 z = 1 holds for complex z; but \|sin z\| and \|cos z\| can exceed 1
- cosh^2 z - sinh^2 z = 1; cos(iy) = cosh y; sin(iy) = i sinh y
- log(-1) = i pi (principal); general: (2n + 1) pi i

3. Limit and Continuity

3.1 Limit

lim f(z) = l as z -> z0 means: for every epsilon > 0 there is a delta > 0 such that |f(z) - l| < epsilon whenever 0 < |z - z0| < delta .

Key difference from real limit

- In real calculus we approach x0 only from left and right (2 ways)
- In complex plane z can approach z0 along infinitely many paths (line, curve, spiral)
- Limit exists only if it is the SAME along all paths ; if two paths give different values -> limit does not exist

Properties

If lim f = l and lim g = m, then lim (f +/- g) = l +/- m ; lim (f g) = l m ; lim (f/g) = l/m (m != 0).

Example 1 - Limit by definition

Show that lim z^2 = 4 as z -> 2. Take z = 2 + h: |z^2 - 4| = |h||h+4| < 5|h| for |h| < 1. Choose delta = min(1, epsilon/5). Done.

Example 2 - Limit does not exist

f(z) = x y / (x^2 + y^2) at z -> 0. Along y = m x: f = m x^2 / (x^2 (1 + m^2)) = m/(1+m^2) - depends on m (slope). Hence limit does not exist .

Example 3

f(z) = z-bar / z at z -> 0. Along real axis (y=0): f = x/x = 1. Along imaginary axis (x=0): f = (-iy)/(iy) = -1. Different -> limit does not exist .

3.2 Continuity

f(z) is continuous at z0 <=> lim f(z) (z -> z0) = f(z0)

i.e. (1) f(z0) defined, (2) limit exists, (3) both are equal. f = u + iv is continuous iff u(x,y) and v(x,y) are both continuous. Sums, products, quotients (denominator != 0) and composites of continuous functions are continuous.

Polynomials, e^z, sin z, cos z are continuous everywhere.

Example: f(z) = (z^2 + 1)/(z - i)... define f(i) = 2i to make it continuous (removable) since z^2+1 = (z-i)(z+i).

4. Differentiation - Derivative of f(z)

$$f'(z_0) = \lim_{h \to 0} \left[f(z_0 + h) - f(z_0) \right] / h$$

The limit must be independent of the way h -> 0 (h complex). If it exists, f is differentiable at z_0. All usual rules (sum, product, quotient, chain) and formulas hold:

f(z)	f'(z)
z^n	n z^(n-1)
e^z	e^z
sin z, cos z	cos z, - sin z
sinh z, cosh z	cosh z, sinh z
log z	1/z
tan z	sec^2 z

Important facts

- Differentiable at a point => continuous there (converse false)
- f(z) = z-bar, |z|^2, Re(z), Im(z) are continuous everywhere but nowhere (or almost nowhere) differentiable
- Complex differentiability is much stronger than real differentiability

5. Cauchy-Riemann Equations

5.1 Derivation

Let $f(z) = u + iv$ be differentiable at $z = x + iy$. Take $h \to 0$ along the real axis ($h = dx$):

$$f'(z) = u_x + i\, v_x$$

Take $h \to 0$ along imaginary axis ($h = i\, dy$):

$$f'(z) = (1/i)(u_y + i\, v_y) = v_y - i\, u_y$$

Both limits must be equal. Comparing real and imaginary parts gives the C-R equations :

Cauchy-Riemann (Cartesian form)

$$u_x = v_y \text{ and } u_y = -v_x$$

Meaning: $u_x = du/dx$ etc. The derivative can then be written $f'(z) = u_x + i\, v_x = v_y - i\, u_y$.

5.2 Polar form of C-R equations

For $z = r\, e^{(i\, theta)}$, $f = u(r, theta) + i\, v(r, theta)$:

$$u_r = (1/r)\, v_{theta} \text{ and } v_r = -(1/r)\, u_{theta}$$

$$f'(z) = e^{(-i\, theta)}\, (\, u_r + i\, v_r\,)$$

5.3 Sufficient condition

If u, v, u_x, u_y, v_x, v_y are continuous in a region and satisfy the C-R equations, then $f(z)$ is differentiable (analytic) there. (C-R equations alone are necessary but not sufficient .)

Example 1 - $f(z) = z^2$

$u = x^2 - y^2$, $v = 2xy$. $u_x = 2x$, $u_y = -2y$, $v_x = 2y$, $v_y = 2x$. So $u_x = v_y$ and $u_y = -v_x$ -> C-R satisfied everywhere, $f'(z) = 2x + i\, 2y = 2z$.

Example 2 - f(z) = z-bar

u = x, v = -y: u_x = 1, v_y = -1 -> u_x != v_y. C-R not satisfied -> not differentiable anywhere.

Example 3 - f(z) = e^z

u = e^x cos y, v = e^x sin y: u_x = e^x cos y = v_y ; u_y = - e^x sin y = - v_x. OK. f'(z) = u_x + i v_x = e^x (cos y + i sin y) = e^z .

Example 4 - f(z) = |z|^2

u = x^2 + y^2, v = 0: u_x = 2x, v_y = 0; u_y = 2y, v_x = 0. C-R holds only at origin (x = y = 0). So differentiable only at z = 0, nowhere analytic .

Example 5 - Show f(z) = sinh z is analytic

sinh z = sinh x cos y + i cosh x sin y. u_x = cosh x cos y = v_y ; u_y = - sinh x sin y = - v_x. C-R satisfied everywhere -> entire .

6. Analytic Functions

6.1 Definition

A function f(z) is analytic (holomorphic, regular) at a point z0 if it is differentiable at z0 and at every point in some neighbourhood of z0 . f is analytic in a region R if it is analytic at every point of R. An entire function is analytic in the whole plane (polynomials, e^z, sin z, cos z).

Differentiable at a point	Analytic at a point
Only at that point is enough	Needs differentiability in a whole neighbourhood
|z|^2 is differentiable at 0	|z|^2 is analytic nowhere
Local property	Stronger property

6.2 Singular points

A point where f(z) fails to be analytic is a singular point . Example: f(z) = 1/(z - 2): singular point z = 2; analytic elsewhere. f(z) = (z + 1)/((z-1)(z+3)): singular at z = 1 and z = -3.

Properties of analytic functions

- Sum, difference, product and composite of analytic functions are analytic, quotient analytic where denominator != 0
- If f is analytic then f'(z), f''(z), ... all exist and are analytic (infinitely differentiable)
- If f = u + iv is analytic, u and v are harmonic
- Curves u = c1 and v = c2 are orthogonal (cut at right angles)
- If f is analytic and f' = 0 everywhere in a region, f is constant
- If f and f-bar are both analytic, f is constant, if |f| is constant, f is constant

Orthogonal trajectories

Slope of u = c1 is $-u_x/u_y$, slope of v = c2 is $-v_x/v_y$. Product = $(u_x\ v_x)/(u_y\ v_y) = (u_x\ v_x)/((-v_x)(u_x)) = -1$ (using C-R). Hence orthogonal.

7. Harmonic Functions

7.1 Laplace equation

A real function phi(x, y) with continuous second partial derivatives is harmonic if it satisfies Laplace equation :

$$\phi_{xx} + \phi_{yy} = 0 \quad (\text{i.e. } \nabla^2 \phi = 0)$$

7.2 Theorem

If f(z) = u + iv is analytic, then u and v are both harmonic. Proof: C-R: $u_x = v_y$, $u_y = -v_x$. Differentiate: $u_{xx} = v_{yx}$, $u_{yy} = -v_{xy}$. Add: $u_{xx} + u_{yy} = v_{yx} - v_{xy} = 0$ (mixed partials are equal). Similarly $v_{xx} + v_{yy} = 0$.

Harmonic conjugate

- If u and v are harmonic and satisfy C-R ($u_x = v_y$, $u_y = -v_x$), then v is the harmonic conjugate of u
- If v is conjugate of u, then u is conjugate of -v (not of v!)
- Not every pair of harmonic functions is a conjugate pair: e.g. $u = x^2 - y^2$ and $v = x^2 - y^2$ are harmonic but not conjugates

Example 1 - Show $u = x^2 - y^2$ is harmonic

$u_x = 2x$, $u_{xx} = 2$; $u_y = -2y$, $u_{yy} = -2$. $u_{xx} + u_{yy} = 0$ -> harmonic .

Example 2 - Show $u = e^x \cos y$ is harmonic

$u_{xx} = e^x \cos y$; $u_{yy} = -e^x \cos y$; sum = 0. Harmonic.

Example 3 - Is $u = x^3 - 3xy^2 + 3x^2 - 3y^2$ harmonic?

$u_{xx} = 6x + 6$, $u_{yy} = -6x - 6$. Sum = 0 -> harmonic .

Example 4 - Is $u = x^2 + y^2$ harmonic?

$u_{xx} + u_{yy} = 2 + 2 = 4 \neq 0$ -> not harmonic . Hence no analytic function has this as real part.

8. Finding the Harmonic Conjugate

8.1 Direct (partial integration) method

1. Verify the given u is harmonic
2. Use C-R: $dv = v_x\,dx + v_y\,dy = -u_y\,dx + u_x\,dy$
3. Integrate (the form is exact since u is harmonic) -> $v = \ldots + C$

$$v = \int (-u_y\,dx + u_x\,dy) \text{ (exact differential)}$$

Example 1

Find the harmonic conjugate of $u = x^2 - y^2 + x$. Here $u_x = 2x + 1$, $u_y = -2y$.

$dv = -u_y\,dx + u_x\,dy = 2y\,dx + (2x + 1)\,dy$. Integrate: $v = 2xy + y + C$

[check: $v_x = 2y$, $v_y = 2x + 1$ OK].

So $f(z) = (x^2 - y^2 + x) + i(2xy + y + C) = z^2 + z + iC$.

Example 2

$u = e^x \cos y$. $u_x = e^x \cos y$, $u_y = -e^x \sin y$. $dv = e^x \sin y\, dx + e^x \cos y\, dy = d(e^x \sin y)$. $v = e^x \sin y + C$, and $f = e^z + iC$.

Example 3 - Given v find u

$v = 3x^2 y - y^3$. $v_x = 6xy$, $v_y = 3x^2 - 3y^2$. $du = u_x\, dx + u_y\, dy = v_y\, dx - v_x\, dy = (3x^2 - 3y^2)\, dx - 6xy\, dy$. Integrating: $u = x^3 - 3xy^2 + C$. $f = z^3 + C$.

9. Construction of Analytic Function -

Given u (or v), find $f(z)$ directly in terms of z without finding the conjugate first.

Milne-Thomson Method

- If u is given: find u_x and u_y. Then $f'(z) = u_x - i\, u_y$ and put $x = z$, $y = 0$
- $f'(z) = u_x(z, 0) - i\, u_y(z, 0)$ -> integrate w.r.t. z: $f(z) = \int [u_x(z,0) - i\, u_y(z,0)]\, dz + C$
- If v is given: $f'(z) = v_y + i\, v_x$ and put $x = z$, $y = 0$
- $f(z) = \int [\, v_y(z,0) + i\, v_x(z,0)\,]\, dz + C$
- C is a complex constant (found from a given condition such as $f(0)$ or $f(1+i)$)

Why does it work?

$x = (z + \bar{z})/2$, $y = (z - \bar{z})/2i$. Since $f(z) = u + iv$ is a function of z only, and identity holds for all $\bar{z}$, put $\bar{z} = z$, i.e. $x = z$, $y = 0$.

Example 1 - $u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$

$u_x = 3x^2 - 3y^2 + 6x$; $u_y = -6xy - 6y$. Put $x = z$, $y = 0$: $u_x = 3z^2 + 6z$, $u_y = 0$.

$f'(z) = u_x - i\, u_y = 3z^2 + 6z$. Integrate: $f(z) = z^3 + 3z^2 + C$. (Constant

u has + 1 so C = 1 + ic.)

Example 2 - u = e^x (x cos y - y sin y)

u_x = e^x (x cos y - y sin y) + e^x cos y ; u_y = e^x (- x sin y - sin y - y cos y).

At (z, 0): u_x = e^z (z + 1) ; u_y = 0. f'(z) = (z + 1) e^z. Integrate by parts: f(z) = z e^z + C.

Example 3 - v = e^x (x sin y + y cos y); find f(z)

v_x = e^x (x sin y + y cos y) + e^x sin y ; v_y = e^x (x cos y + cos y - y sin y). At (z,0): v_x = 0, v_y = e^z (z + 1).

f'(z) = v_y + i v_x = e^z (z + 1) -> f(z) = z e^z + C .

Example 4 - u - v given (important type)

If u - v = (x - y)(x^2 + 4xy + y^2), find f(z) = u + iv. Let F = (1+i) f = (u - v) + i (u + v). Then U = u - v (real part of analytic F). U_x = 3x^2 + 6xy - 3y^2... computing: U = x^3 + 3x^2 y - 3x y^2 - y^3 ; U_x = 3x^2 + 6xy - 3y^2 ; U_y = 3x^2 - 6xy - 3y^2. At (z,0): U_x = 3z^2, U_y = 3z^2. F' = U_x - i U_y = 3(1 - i) z^2 -> F = (1 - i) z^3 + C. So f = F/(1+i) = ((1-i)/(1+i)) z^3 = -i z^3 + C' .

Given	Use	Formula
u	Milne-Thomson	f' = u_x - i u_y at (z, 0)
v	Milne-Thomson	f' = v_y + i v_x at (z, 0)
u + v or u - v	Consider (1 + i) f or (1 - i) f	(1+i) f = (u - v) + i (u + v)
Polar u(r,theta)	Use polar C-R	f' = e^(-i theta)(u_r + i v_r), put r = z, theta = 0

Quick revision - Unit I

- $w = f(z) = u + iv$, limit must be same along all paths
- C-R: $u_x = v_y$, $u_y = -v_x$; polar: $u_r = v_\theta / r$, $v_r = -u_\theta / r$
- Analytic = differentiable in a neighbourhood, u, v harmonic: $u_{xx} + u_{yy} = 0$
- Harmonic conjugate: $v = \int(-u_y\,dx + u_x\,dy)$
- Milne-Thomson: $f' = u_x - i\,u_y$ $(x = z, y = 0)$ or $v_y + i\,v_x$